CYCLIC INTEGRATION TECHNIQUES

Derivation and Application of Generalized Formulas for Exponential-Trigonometric Products

In integral calculus, a class of functions known as **cyclic integrands** does not terminate under repeated differentiation or integration. When evaluating products of exponential and trigonometric functions—such as $e^{ax}\sin(bx)$—applying Integration by Parts twice reproduces a scaled version of the original integral. This property allows us to isolate and solve for the integral algebraically.

1. GENERAL THEOREM & CLOSED-FORM EXPRESSIONS

For any non-zero real constants a and b, the general reduction formulas for exponential-trigonometric cyclic integrals are given by:

$$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}(a\sin(bx) - b\cos(bx))}{a^2 + b^2} + C$$

$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$$

2. DERIVATION VIA DOUBLE INTEGRATION BY PARTS

Let $I = \int e^{ax}\sin(bx)\,dx$. We apply Integration by Parts, defined as $\int \mathbf{u\,dv = uv - \int v\,du}$.

Step 1: First Integration by Parts

Let $\mathbf{u = \sin(bx)} \Rightarrow \mathbf{du = b\cos(bx)\,dx}$

Let $\mathbf{dv = e^{ax}\,dx} \Rightarrow \mathbf{v =} \frac{\mathbf{e^{ax}}}{\mathbf{a}}$

Applying the formula yields:

$$I = \frac{1}{a}e^{ax}\sin(bx) - \frac{b}{a}\int e^{ax}\cos(bx)\,dx$$

Step 2: Second Integration by Parts

For the remaining integral $\int e^{ax}\cos(bx)\,dx$, assign:

Let $\mathbf{u = \cos(bx)} \Rightarrow \mathbf{du = -b\sin(bx)\,dx}$

Let $\mathbf{dv = e^{ax}\,dx} \Rightarrow \mathbf{v =} \frac{\mathbf{e^{ax}}}{\mathbf{a}}$

Substituting back into the auxiliary integral:

$$\int e^{ax}\cos(bx)\,dx = \frac{1}{a}e^{ax}\cos(bx) - \int \frac{1}{a}e^{ax}(-b\sin(bx))\,dx = \frac{1}{a}e^{ax}\cos(bx) + \frac{b}{a}\int e^{ax}\sin(bx)\,dx$$

3. APPLICATION MATRIX FOR CYCLIC FAMILIES

The cyclic method applies to several standard pairings. The general approach remains identical: apply integration by parts until the original structure reappears, then perform algebraic collection.

Integrand Structure	Recurrence Pattern	Solving Mechanism
e^{ax} sin(bx) or **e^{ax} cos(bx)**	Reappears after 2 steps with coefficient $-b^2/a^2$	Add (b^2/a^2)I to left side; multiply by inverse factor.
sinh(ax) sin(bx)	Reappears after 2 steps with coefficient $+b^2/a^2$	Algebraic collection based on hyperbolic derivative identities.
sin(ax) cos(bx)	Reappears after 2 steps (or simplified via product-to-sum)	Direct linear reduction or standard trigonometric identities.
$\sec^3(x)$	Reappears as $\int \sec(x)\,dx$ after identity transformation	Use $\sec^2(x) = 1 + \tan^2(x)$; collect $2\int \sec^3(x)\,dx$.

4. WORKED CONCRETE EXAMPLES

EXAMPLE 1: PARAMETER EVALUATION (A = 3, B = 4)

Evaluate $\int e^{3x} \sin(4x)\,dx$ using the general formula.

Solution: Identify constants $a = 3$ and $b = 4$. Substitute directly into the general reduction formula:

$$\int e^{3x} \sin(4x)\,dx = \frac{e^{3x}(3\sin(4x) - 4\cos(4x))}{3^2 + 4^2} + C = \frac{e^{3x}(3\sin(4x) - 4\cos(4x))}{25} + C$$

EXAMPLE 2: COSINE CYCLIC VARIANT (A = 2, B = 1)

Evaluate $\int e^{2x} \cos(x)\,dx$.

Solution: Here $a = 2$ and $b = 1$. Applying the general cosine formula:

$$\int e^{2x} \cos(x)\,dx = \frac{e^{2x}(2\cos(x) + 1\sin(x))}{2^2 + 1^2} + C = \frac{e^{2x}(2\cos(x) + \sin(x))}{5} + C$$

5. KEY TAKEAWAYS & CONDITIONS

• **Consistency is mandatory:** Choose the exponential term as dv (or u) consistently in both integration steps. Reversing choices on step 2 will cause terms to cancel out, resulting in $0 = 0$.

• **Sign checking:** Double-check negative signs when differentiating cosine ($d/dx\,[\cos(bx)] = -b\sin(bx)$), as sign errors prevent correct algebraic collection.

• **Alternative derivation:** These formulas can also be derived concisely using Euler's identity: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ by evaluating the real and imaginary parts of $\int e^{(a+ib)x}\,dx$.